

Learning Peak Temperature in 3DICs by Deep Differentiable Forest

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Abstract—Machine learning is a very attractive idea for thermal problems from 3D integrated circuit. This paper study the peak temperature problem under various power configurations. We give the detailed model and algorithm of deep differentiable forest. A complex 3DIC device with hybrid bonding and heterogeneous technology nodes integration is used to test this model. Commercial finite element tool needs million grids to get high accuracy solution. Compare to FEM tool, our model is hundreds of times faster and the relative error is less than 1%. We also compare its performance with GBDT and random forest model. The experimental results show differentiable forest is a model with higher accuracy in predicting peak temperature.

Index Terms—3D ICs, Peak temperature, Thermal simulation, Machine learning, Differentiable forest

I. INTRODUCTION

A thermal simulation of 3D IC system may involve multiple components with different technology nodes. In this paper, we build a 3DIC model involves three dies and an interposer as shown in the following Figure 1:

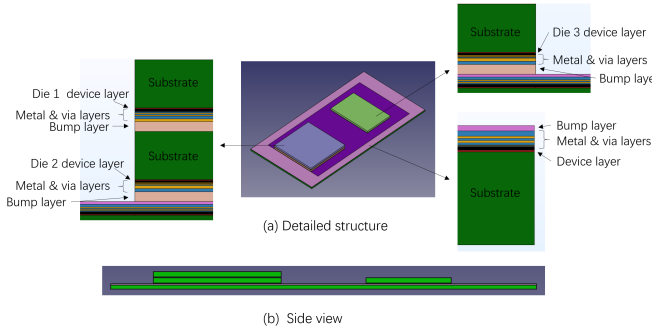


Fig. 1: Detailed structure of 3DIC model involves three dies and an interposer

DIE1 and DIE2 have 14nm equivalent technology while DIE3 and interposer have 16nm equivalent technology. DIE1 and DIE2 are connected by hybrid bonding with face-to-back style. DIE2 and DIE3 are separately connected to interposer by micro bump with side-by-side style. The model in this paper contains several key characteristics of state-of-the-art chiplet designs including:

- Direct die to die connection by hybrid bonding
- Heterogeneous technology nodes integration

The interposer size is $2.5 \times 5 \text{ mm}$, and the die size are $1.5 \times 1.5 \text{ mm}$, $1.5 \times 1 \text{ mm}$ respectively. Both die and interposer are constructed of multi-layer structures, including metal and via layers, device layer, substrate layers. All dies are flipped so that the substrate layers are on the top.

II. EXPERIMENT

To get experiments datasets, we generate 20,000 random power maps by some rules. For each power map, peak temperature is from the simulation result of a commercial FEM tool. Each case needs 219 seconds (on 8-threads Intel Xeon Platinum 8356H 32C/64T@3.9GHz).

We compare differentiable forest model with two other classic decision tree models: random forest and gradient-boosted decision trees(XGBoost).

We measure the accuracy of prediction value by relative error:

$$err = \frac{abs(\Gamma - \hat{T})}{\Gamma} \quad (1)$$

where Γ is result of commercial FEM software and $\hat{T}$ is the prediction of learned model. As table I shows, the relative error of differentiable forest is only 0.11%. It's lower than the relative error of XGBoost(0.28%) and random forest(0.4%). This is also confirmed by Figure 2, which shows the peak temperature distribution of each model.

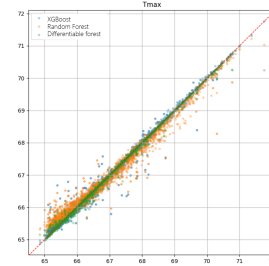


Fig. 2: The peak temperature distribution of three models

Table I list the training time of each model. So our model is much faster than random forest and nearly same as XGBoost.

TABLE I: Comparison of three models

Model	Relative error(%)	Training time(sec)
Differentiable forest	0.11%	116.3
XGBoost	0.28%	109.1
Random forest	0.4%	1891.7

Differentiable forest model needs less than 1 second to predict peak temperature from each power map. Compare to 219 seconds by commercial FEM tool, the speedup of this model exceeds 200x!

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